QUESTION 4 [15 MARKS]

Figure 2 shows two processes, Process A and Process B, that share the same variable x initialized to x = 100. Process A wants to increment the variable x, while Process B wants to decrement the variable x. Both run in an infinite loop.

```
do {
      acquire();
      x += 1
      release();
}while(True)//process A
```

Process A

```
do {
      acquire();
      x -= 1
      release();
}while(True); //ProcessB
```

Process B

Figure 2

a) Complete the definition of the acquire() and release() function as shown in the Figure 3 below (for items labelled as **a** and **b**). [2 marks]

```
acquire() {
      while(!available);
      _____ ( a ) ____;
}
```

acquire()

```
release() {
      _____ ( b ) ____;
}
```

release()

Figure 3

(a) available = false (b) available = true

b) Assume that the CPU scheduler schedules the processes alternately, i.e., Process A, Process B, Process A, Process B. Continue trace the instruction of the processes as well as the value of `x` and `available` in the Table 4 below until t=10. [5 marks]

Copy Table 4 to your answer booklet to answer the question:

time	Process A	Process B	Available	X
T0	Acquire()	Acquire()	True	X=100
T1	X+=1	Acquire()	False	X=101
T2	Release()	Acquire()	True	X=101
…				
T10				

Table 4

time	Process A	Process B	Available	X
T0	acquire()	acquire()	true	x=100
T1	x+=1	acquire()	false	x=101
T2	release()	acquire()	true	x=101
T3	acquire()	x-=1	false	x=100
T4	acquire()	release()	true	x=100
T5	x+=1	acquire()	false	x=101
T6	release()	acquire()	true	x=101
T7	acquire()	x-=1	false	x=100
T8	acquire()	release()	true	x=100
T9	x+=1	acquire()	false	x=101
T10	release()	acquire()	true	x=101

c) Modify the code in Figure 2 to use semaphores using the correct semaphore operations. [3 marks]

`do {`
 	
`wait(S);`
`x+=1;`
`signal(S);`
 	
`}while(true);`
	`do {`
	
	`wait(S);`
	`x-=1;`
	`signal(S);`
	
	`}while(true);`

Process A	**Process B**

d) Using the semaphore algorithm, trace the execution of Process A and Process B (as in Question 4 (a) and (b)) by stating the instructions executed by Process A and B as well as the value of X and the integer semaphore S. Set the initial condition of S=1 and the processes execute alternately beginning with Process A. [5 marks]

Copy Table 5 to your answer booklet to answer the question:

Time	Process A	Process B	S	X
T0	wait()	wait()	1	100
T1	x+=1	wait()	0	101
T2				
…				
T10				

Table 5

Time	Process A	Process B	S	X
T0	wait()	wait()	1	100
T1	x+=1	wait()	0	101
T2	signal()	wait()	1	101
T3	wait()	x-=1	0	100
T4	wait()	signal()	1	100
T5	x+=1	wait()	0	101
T6	signal()	wait()	1	101
T7	wait()	x-=1	0	100
T8	wait()	signal()	1	100
T9	x+=1	wait()	0	101
T10	signal()	wait()	1	101